

Project Proposal: IoT-Based Temperature Monitoring System

Project Title: IoT-Based Temperature Monitoring System using Artix 7, PMOD TMP 2, PMOD OLED RGB, ESP32, and Active Buzzer

Project Overview:

The aim of this project is to develop an Internet of Things (IoT)-based system that monitors the room temperature and displays the data on an OLED screen. Simultaneously, the temperature readings will be sent to a mobile platform via the ESP32 wireless module. The system will also trigger an active buzzer if the temperature exceeds a predefined threshold, indicating an alert.

Objectives:

1. Measure the room temperature using the PMOD TMP 2 temperature sensor.
2. Display the temperature on a PMOD OLED RGB screen in real-time.
3. Transmit the temperature data via the ESP32 to a mobile platform (Android/iOS) for remote monitoring.
4. Trigger an active buzzer when the temperature exceeds the set limit, acting as a warning signal.

Components Used:

- **Artix 7 100T:** FPGA to process the data from the temperature sensor.
- **PMOD TMP 2:** Temperature sensor to detect the room temperature.
- **PMOD OLED RGB:** OLED display to show the temperature readings.
- **ESP32:** Wireless communication module to send the temperature data to a mobile platform.
- **Active Buzzer:** To alert when the temperature exceeds a threshold.

Methodology:

1. **Sensor Integration:** The PMOD TMP 2 will be connected to the Artix 7 FPGA, which will read the temperature data.

2. **Data Display:** The temperature data will be displayed in real-time on the PMOD OLED RGB screen.
3. **Wireless Communication:** The ESP32 will handle the transmission of temperature data to a mobile platform using Wi-Fi, enabling remote monitoring of the room temperature.
4. **Alert Mechanism:** An active buzzer will be triggered if the temperature exceeds a predefined threshold, alerting users to potential overheating.

Expected Outcome:

The system will provide real-time temperature monitoring with local and remote data display. It will also offer an alert mechanism for overheating situations, enhancing room temperature control and safety.